

EXPERIMENT 9

Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

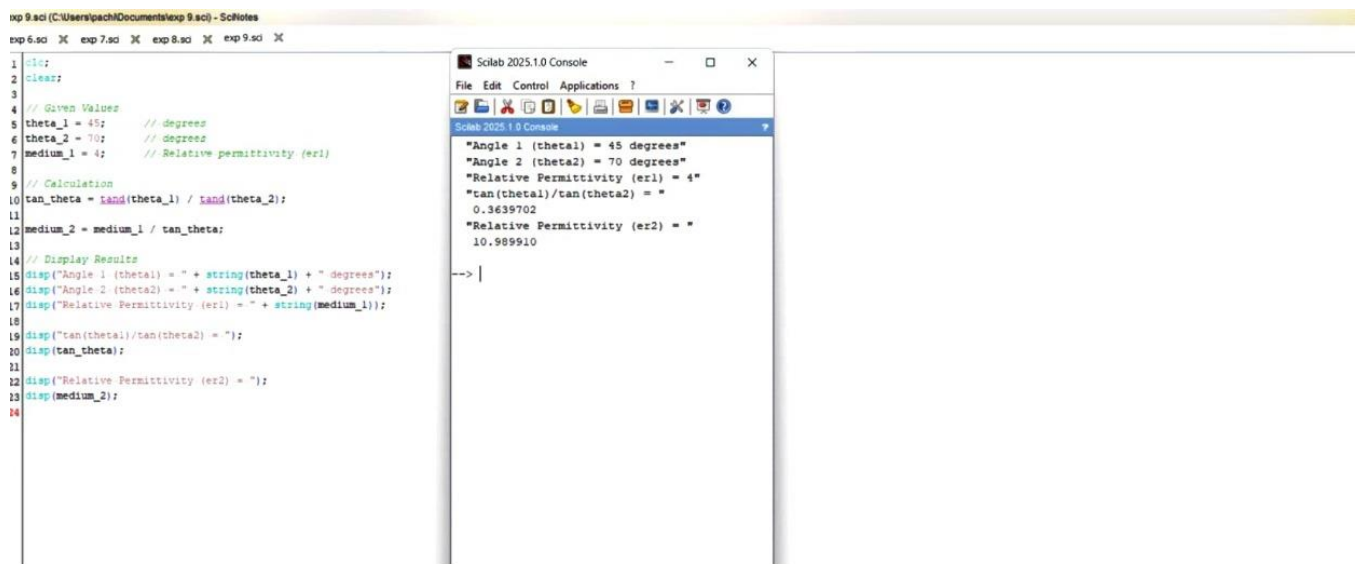
Aim:

To write program for Dielectric – free space boundary condition in Electric field using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:



```
1 clear;
2 clear;
3
4 // Given Values
5 theta_1 = 45; // degrees
6 theta_2 = 70; // degrees
7 medium_1 = 4; // Relative permittivity (er1)
8
9 // Calculation
10 tan_theta = tand(theta_1) / tand(theta_2);
11
12 medium_2 = medium_1 / tan_theta;
13
14 // Display Results
15 disp("Angle 1 (theta1) = " + string(theta_1) + " degrees");
16 disp("Angle 2 (theta2) = " + string(theta_2) + " degrees");
17 disp("Relative Permittivity (er1) = " + string(medium_1));
18
19 disp("tan(theta1)/tan(theta2) = ");
20 disp(tan_theta);
21
22 disp("Relative Permittivity (er2) = ");
23 disp(medium_2);
24
```

Scilab 2025.1.0 Console

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Scilab 2025.1.0 Console

"Angle 1 (theta1) = 45 degrees"
"Angle 2 (theta2) = 70 degrees"
"Relative Permittivity (er1) = 4"
"tan(theta1)/tan(theta2) = "
0.3639702
"Relative Permittivity (er2) = "
10.989910

--> |

Theoretical calculations:

$$\theta_{ap} = 9$$

$$\theta_1 = 45^\circ$$

$$\theta_2 = 70^\circ$$

$$e_1 = 4$$

calculate tangent

$$\tan 45^\circ = 1$$

$$\tan 70^\circ = 2.7475$$

calculate ratio

$$\tan \theta = \frac{1}{2.7475}$$

$$\tan \theta = 0.36397$$

$$\Sigma \pi_2$$

$$\Sigma \pi_2 = \frac{4}{0.36397}$$

$$\boxed{\Sigma \pi_2 = 10.99}$$

Case 1: In this case, the dielectric medium has a relative permittivity (ϵ_r) of 2, meaning the

Dielectric reduces the electric field by half compared to the free space. This is typical for

Materials like mica or certain types of plastics.

- Free Space Electric Field (E_1): 100 V/m
- Dielectric Electric Field (E_2): 50 V/m

Case 2: Here, the dielectric medium has a higher relative permittivity ($\epsilon_r=2.5$) compared to

Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like

Glass or water.

- Parameters:

○ $E_{n1}=200$ V/m (Electric field in free space).

○ $E_{n2}=80$ V/m (Electric field in the dielectric medium).

Case 3: The dielectric medium here also has a relative permittivity of 2. The behavior of the

Material suggests it could be another low-polarization dielectric material, with a consistent

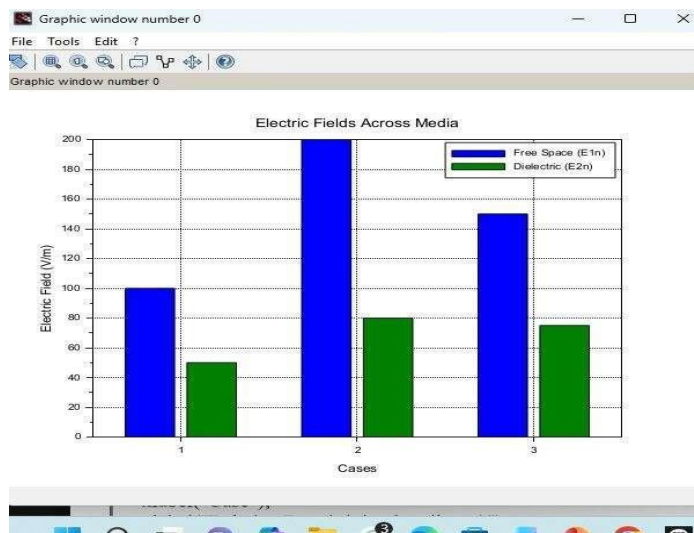
Reduction of the electric field by half.

- Free Space Electric Field (E_{1n}): 150 V/m
- Dielectric Electric Field (E_{2n}): 75 V/m

Program:

```
EXP 9.sci (C:\Users\user\EXP 9.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 9.sci (C:\Users\user\EXP 9.sci) - SciNotes
EXP 9.sci X
3 clear
4 clf
5
6 //Constants
7 epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
8
9 //Case Parameters
10 E1n = [100 200 150]; // Electric field in free space (V/m)
11 E2n = [50 80 75]; // Electric field in dielectric (V/m)
12
13 //Compute Relative Permittivity
14 epsilon_r = E1n ./ E2n;
15
16 //Display Results
17 disp("Relative Permittivity for Each Case:");
18 disp(epsilon_r);
19
20 //Case numbers
21 n = [1 2 3];
22
23 //-----
24 // Bar Graph
25 //-----
26 h = figure;
27 bar(n, [E1n' E2n']);
28
29 xlabel("Cases");
30 ylabel("Electric Field (V/m)");
31 title("Electric Fields Across Media");
32 legend(["Free Space (E1n)", "Dielectric (E2n)"]);
33 axis([1 3 0 200]);
34
35 //-----
36 // Line Graph
37 //-----
38 h = figure;
39 plot(n, epsilon_r, "-o");
40
41 xlabel("Cases");
42 ylabel("Relative Permittivity");
43 title("Relative Permittivity for Each Case");
44 axis([1 3 0 2]);
45
46 //Display values
47 disp("Case-1 ER = " + string(epsilon_r(1)));
48 disp("Case-2 ER = " + string(epsilon_r(2)));
49 disp("Case-3 ER = " + string(epsilon_r(3)));
50
```

Output:



Theoretical calculations:

1. Case 1:

$$\varepsilon_r = \frac{E_{1n}}{E_{2n}} = \frac{100}{50} = 2$$

2. Case 2:

$$\varepsilon_r = \frac{E_{1n}}{E_{2n}} = \frac{200}{80} = 2.5$$

3. Case 3:

$$\varepsilon_r = \frac{E_{1n}}{E_{2n}} = \frac{150}{75} = 2$$

Result:

Thus the program was verified for Dielectric – free space boundary condition in Electric field using SCILAB was successfully.